Mathematics in LaTeX: Beginner's Guide

Contents

1 Inline and Display Mathematics

Definition. LaTeX has two common ways to write mathematical expressions. Inline mathematics sits within a sentence. Display mathematics appears separately, usually centered on its own line.

Example 1: Inline mathematics

```
The equation $a+b=c$ is an example.
```

Expected result: The equation $a+b=c$ is an example.

Example 2: Display mathematics

```
\[
a+b=c
\]
```

Expected result:

$$a+b=c$$

Commands. Dollar signs `$...$` mark inline maths. The commands `\[` and `\]` begin and end an unnumbered display equation.

Practice. Write $x+y=z$ inside a sentence and then on its own line.

2 Superscripts and Subscripts

Definition. A superscript is placed above a symbol, often to show a power. A subscript is placed below a symbol, often to label it. **Source code:**

```
\[ x^2 \qquad a_1 \qquad x_1^2 \]
\[ x^{10} \qquad a_{12} \]
```

Expected result:

$$x^2 \qquad a_1 \qquad x_1^2$$

$$x^{10} \qquad a_{12}$$

Commands. The caret `^` creates a superscript; the underscore `_` creates a subscript. Use braces to group more than one character, as in `x^{10}`. **Practice.** Write y^3, b_2, and m_{15}.

3 Fractions and Roots

Definition. A fraction has a numerator (top part) and denominator (bottom part). A root shows a value that, when raised to a power, gives the original number. **Source code:**

```
\[ \frac{a}{b} \qquad \sqrt{x} \qquad \sqrt[3]{x} \]
```

Expected result:

$$\frac{a}{b} \qquad \sqrt{x} \qquad \sqrt[3]{x}$$

Commands. `\frac{a}{b}` creates a fraction. `\sqrt{x}` creates a square root. The optional index in `\sqrt[3]{x}` makes a cube root. **Practice.** Write one-half, the square root of $a + b$, and the fourth root of x.

4 Greek Letters

Definition. Greek letters are commonly used for mathematical quantities, angles, and constants. **Source code:**

```
\[ \alpha,\ \beta,\ \gamma,\ \delta \]
\[ \pi,\ \theta,\ \lambda,\ \omega \]
```

Expected result:

$$\alpha,\ \beta,\ \gamma,\ \delta$$

$$\pi,\ \theta,\ \lambda,\ \omega$$

Commands. A backslash followed by a letter name produces the symbol: `\alpha` gives α, and `\theta` gives θ. The command `\pi` gives π. **Practice.** Find and write the commands for μ, σ, ϕ, and Δ.

5 Mathematical Operators

Definition. Operators represent mathematical actions. Summation adds terms, product multiplies terms, and a limit describes a value an expression approaches. **Source code:**

```
\[ \sum_{i=1}^{n} i \]
\[ \prod_{i=1}^{n} i \]
\[ \lim_{x \to 0} f(x) \]
```

Expected result:

$$\sum_{i=1}^{n} i$$

$$\prod_{i=1}^{n} i$$

$$\lim_{x \to 0} f(x)$$

Commands. `\sum` makes a summation sign, `\prod` makes a product sign, and `\lim` makes a limit operator. Underscores and carets set the lower and upper limits. **Practice.** Write a summation from $i = 1$ to 5 and a limit as x approaches 2.

6 Brackets and Grouping

Definition. Brackets group parts of an expression. LaTeX can also resize brackets to fit tall expressions such as fractions. **Source code:**

```
\[ (a+b) \qquad [x+y] \]
\[ \left(\frac{a}{b}\right) \]
\[ \left[\frac{x+1}{x-1}\right] \]
```

Expected result:

$$(a+b) \qquad [x+y]$$

$$\left(\frac{a}{b}\right)$$

$$\left[\frac{x+1}{x-1}\right]$$

Commands. Type ordinary parentheses and square brackets directly. Pair `\left` with `\right` to resize matching delimiters around an expression. **Practice.** Put large round brackets around a fraction and large square brackets around a sum.

7 Equations and Alignment

Definition. Displayed equations appear on separate lines. An alignment environment lines up related equations at a chosen point, commonly the equals sign. **Source code:**

```
\begin{align*}
x + 2 &= 5 \\
x &= 5 - 2 \\
x &= 3
\end{align*}
```

Expected result:

$$\begin{aligned} x + 2 &= 5 \\ x &= 5 - 2 \\ x &= 3 \end{aligned}$$

Commands. The `align*` environment (from the `amsmath` package) aligns multiple lines without numbering them. The ampersand `&` marks the alignment point, and `\\` starts a new row. Add `\usepackage{amsmath}` to the document preamble. **Practice.** Show the steps for solving $y+4=9$, aligning each equals sign.

8 Matrices

Definition. A matrix is a rectangular arrangement of entries in rows and columns. **Source code:**

```
\[
\begin{bmatrix}
1 & 2 \\
3 & 4
\end{bmatrix}
\]
```

Expected result:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Commands. The `bmatrix` environment creates a matrix with square brackets. Use `&` to separate columns and `\\` to start a new row. It is provided by the `amsmath` package. **Practice.** Create a matrix with two rows and three columns containing the numbers 1 to 6.